Budgeted Model Selection for Weakly Supervised Knee MRI

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Project: AYMENsl25/RSNA-competion-kaggle-project-. This retrospective report documents the V2 study and its later competition experiments. It has not been peer reviewed. Human author names and affiliations are not assigned by this document. AI assistance was used in implementation, analysis and report preparation.

Abstract

We examine model selection for twelve-finding knee MRI classification when most supervision is derived from radiology reports and compute is limited. We developed a shared-encoder 2.5D pipeline, investigated pretraining, pooling, series coverage, context and label processing, and deployed a four-model ensemble. That independently trained ensemble achieved 0.857 public and 0.837 private competition macro AUC. A separate reproduction of an existing public checkpoint stack achieved 0.958 and 0.942, respectively. Under an early three-epoch protocol, ImageNet initialization increased proxy AUC from 0.7397 to 0.8017, while denser training windows and two-series sampling did not improve the matched control. In a later comparison, attention pooling improved proxy AUC by 0.0123 in one seed, but its advantage did not reproduce in a second matched pair. Frozen DINOv2 experiments showed a combined context/capacity/metadata gain of 0.0186, while a matched-capacity, post hoc control isolated a much smaller geometry increment of 0.0014. An exploratory ranking-loss arm improved proxy AUC by 0.0089. These findings remain conditional on weak references, limited seeds and reused validation data. We describe provenance, resource costs, numerical failures and the distinction between improved proxy agreement and improved hidden-test performance.

1. Task and study scope

The RSNA Knee Abnormality Detection competition required twelve study-level MRI predictions: ACL, MCL, medial/lateral meniscus, medial/lateral/patellofemoral OA, effusion, synovitis, Baker's cyst, contusion and fracture. The official development cohort contained 4,407 studies and 24,371 series, with expert labels for 58 studies. Most supervision therefore had to be constructed from reports or other permitted sources. Evaluation used mean per-finding ROC-AUC. Test-time readers received MRI without report text; submission inference ran offline [1].

Our study has two tracks. The independently trained track develops readers from generic ImageNet or DINO initialization. The public-weights track reproduces and adapts competition checkpoints released by other participants. Combining their scores into a single architecture claim would obscure their different provenance. The results below are reported by track and by evaluation reference.

This is a retrospective account of adaptive experimentation. Some comparisons had fixed endpoints and promotion rules before execution; the entire study was not prospectively registered. Plans, completed runs, exploratory diagnostics and deployment measurements are distinguished throughout.

2. Data, supervision and evaluation cohorts

2.1 Report proxies and missing targets

Early experiments used a frozen public report-probability source. Observed probabilities above or below 0.5 became positive or negative targets; exactly 0.5 was masked. Later experiments retained soft probabilities and used clipped positive-class loss weights. Report mentions and report silence do not establish the competition's image-positive criteria. AUC against these targets measures proxy agreement rather than clinical diagnostic accuracy.

The early report-v3 label process applied training-developed severity rules to effusion and Baker's cyst: qualifying mild-only mentions could be reduced to a soft target of 0.3. It also retained soft values and masked unknown cells. These

changes were deployed together with loss changes in E4+E5; their individual leaderboard contributions cannot be separated from that submission comparison.

A different, later teacher-v3 process blended report consensus with grouped out-of-fold image-teacher predictions. The historical name "v3" appears in both tracks; report-v3 and teacher-v3 are not interchangeable target files. This report does not publish either label table.

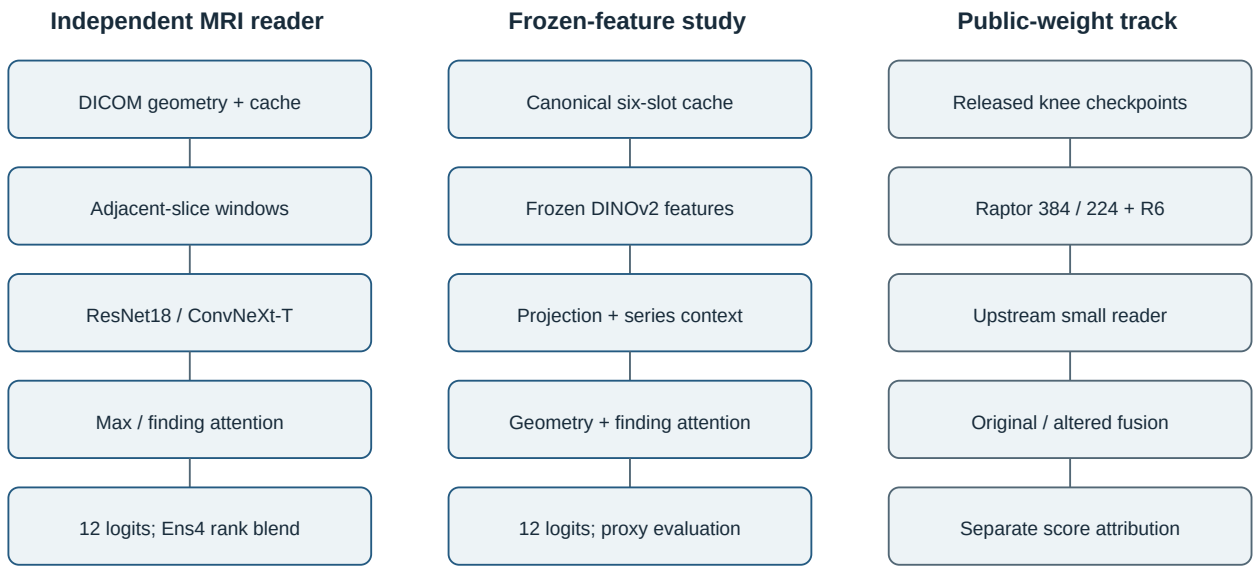
2.2 Cohort versions

Panel	Eligible training	Validation	Known validation groups	Reference
Early M/P/D/S comparisons	3,428	863	839	Frozen report proxy
Later N-series comparisons	3,428	885	843	Frozen public/v3 proxy tables
F2/F4 frozen-feature study	3,427	863	839	Frozen public report proxy
Competition submissions	Model-specific training	Hidden test	Unavailable locally	Organizer image labels

Known patient/report links and duplicate-audit exclusions informed grouping. This does not prove complete patient or site independence. Early exclusions included 65 expert-connected studies. Different all-masked-target rules explain part of the validation-count difference between panels. Cohort membership, reference versions and actual valid-target masks must accompany any recomputation.

F2 excluded one training study after selected-series cache QC failed; it used no replacement series and had no validation QC exclusion. The expert cases were excluded from its fitting. Later public-weight adaptation studies used small expert-set diagnostics, and upstream competition checkpoints had their own training and expert-selection histories. Consequently, the expert58 cannot be described as untouched across the whole project.

3. Architecture and preprocessing



Separate pipelines and endpoints; schematic, not patient imagery.

Figure 1. The independent reader and the frozen-feature experiment are separate pipelines. All diagrams are original schematics; no patient images or measured attention maps are shown.

3.1 Canonical cache and sampling in the Ens4 track

The V3 preprocessing function orders DICOM slices by physical position projected onto the slice normal. It canonicalizes in-plane axes with flips/transposes, forms a central 160-mm crop or pad, resizes to 256 pixels and scales intensities using foreground percentiles before uint8 caching. Laterality is recorded in the cache and applied by the loading/inference path. Mirroring reverses sagittal slice order or flips coronal/axial columns, preserving the intended medial/lateral meaning. The same canonicalization implementation is shared with inference.

For the three-slot readers, one series per sagittal/coronal/axial role is selected using the fluid-sensitive, fat-suppression and UID preference. A window contains three adjacent source slices from a single series. Most training recipes sample four random windows per role; inference uses sixteen evenly spaced windows per role. Inputs are resized to 224 pixels and normalized with ImageNet statistics. The E2 variant uses fixed slab sampling. Missing windows are masked and do not enter the encoder or pooling operation.

These statements describe the Ens4 preprocessing generation. The early baseline and later native-resolution caches have their own frozen source versions; they must not be silently replaced with this cache in a purported reproduction.

3.2 Shared encoder and aggregation

ResNet18 or ConvNeXt-Tiny maps each valid 2.5D window to a feature vector. The max head takes the feature-wise maximum over windows and applies a twelve-output linear layer. The attention head learns a separate weighting over windows for each finding, with gated attention and a plane bias, following established attention-MIL principles [2].

For a study with window features h and finding c, the general attention form is:

```
alpha[k,c] = softmax_over_valid_windows(attention_c(h[k]))
pooled[c]  = sum_k alpha[k,c] * h[k]
logit[c]   = dot(classifier[c], pooled[c]) + bias[c]
```

The six-slot bridge distinguishes plane and fluid/non-fluid contrast. The N3b context package adds a 384-dimensional projection, learned slot and normalized slice-position embeddings, and two Transformer layers before finding-specific pooling. This package changes several components together; its outcome cannot be assigned solely to self-attention.

3.3 Independently trained deployed ensemble: Ens4

Member	Backbone	Aggregation	Main recipe differences
E4+E5	ImageNet ResNet18	Max	Report-v3 soft targets and clipped positive weighting
E6	ImageNet ResNet18	Max	Anatomical augmentation, 16 epochs, checkpoint averaging
E2	ImageNet ResNet18	Finding attention	Fixed slab sampling
E8	ImageNet ConvNeXt-Tiny	Max	Different backbone and optimizer recipe

Ens4 averages per-finding percentile ranks of these four members with equal weights. Rank conversion is performed over the evaluation population, making the final values cohort-relative scores rather than calibrated probabilities. Individual members predict MRI findings without report input at inference.

Typical fitting used batch size two, AdamW, cosine decay and fp16 AMP. ResNet recipes used learning rate 3e-4 and weight decay 0.01; the ConvNeXt-Tiny recipe used 1e-4 and 0.05. Training lasted 12 epochs for most members and 16 for E6. The reported deployment bundle retains its selected/averaged checkpoints and does not represent four otherwise identical models.

3.4 Frozen DINOv2 experiment

F2 encodes 224-pixel adjacent-slice windows using a frozen generic DINOv2 ViT-S/14 encoder. Up to eight deterministic centers are retained in each of six plane/contrast slots. The 384-dimensional CLS feature is projected to 128 dimensions for CPU head training. DINO plus sequence aggregation already has medical-volume precedent, including knee MRI [3].

Head A uses finding-specific pooling. Head B adds slot embeddings, five physical/relative geometry values and a within-series Transformer. Geometry includes centered physical position, neighboring gaps, series extent and relative position. Cross-series registration is not assumed. Head C retains B's capacity and initialization construction but receives zero geometry; this diagnostic was added after the initial A/B result.

F4 adds two separate CPU experiments to B: BCE plus a 0.25-weighted pairwise logistic ranking term, and whole-series training dropout with probability 0.2. The dropout arm retains at least one available series. These use the same features, training order and twelve-epoch schedule, with independent seeded dropout draws. Ranking is an elementary surrogate, not a reproduction of a specialized deep-AUC optimizer.

4. Statistical and engineering protocol

The primary development measure is twelve-target macro ROC-AUC on the specified proxy table. Saved comparisons use paired resampling of known groups, commonly 1,000 draws; F4 uses 500. The same resampled groups are used for both models in a comparison. These intervals measure evaluation-sample variation conditional on a fitted run and reference. They do not capture training-seed variance, systematic label error, multiple comparisons or adaptive selection.

Some comparisons fix the final epoch; F2/F4 select highest observed validation AUC, then lower BCE, then earlier epoch. These selected-checkpoint intervals are exploratory because the same fold is used for selection. Equal epochs also do not imply equal optimizer updates, examples or compute. An altered cosine horizon changes the optimization trajectory.

Source/configuration hashes, split and label hashes, saved predictions, output schemas, masks, failures and resource measurements are retained. Selected F2 checkpoints have historical strict-reload receipts reproducing all 863 by 12 validation probabilities exactly. Report preparation checked the receipt and source hashes; it did not repeat training or recompute all per-study metrics.

The report package contains aggregate tables only. The publication evidence manifest records source-document and aggregate-artifact hashes without exposing study identifiers, reports, labels, predictions or model weights.

5. Results

5.1 Initialization and coverage at three epochs

Variant	Proxy macro AUC	Script hours	Peak reserved MiB
M00 random-init ResNet18, max	0.739719	2.62	760
P00 ImageNet ResNet18, max	0.801735	2.39	760
D01 doubled training windows	0.793934	3.24	982
S01 two series/plane, fixed windows	0.792733	4.09	760

P00 minus M00 was +0.062016, with a paired group interval [0.049379, 0.075216]. D01 and S01 were below the matched P00: -0.007801 [-0.014713, -0.001354] and -0.009002 [-0.017304, -0.001241], respectively. This supports ImageNet initialization in this protocol, but not a general claim that additional series are harmful. S01 increases decoding work despite a fixed encoder window budget. The separate six-epoch E01 reached 0.821538 and required 5.10 script hours; duration and schedule differ from the three-epoch controls.

5.2 Attention replication and the context package

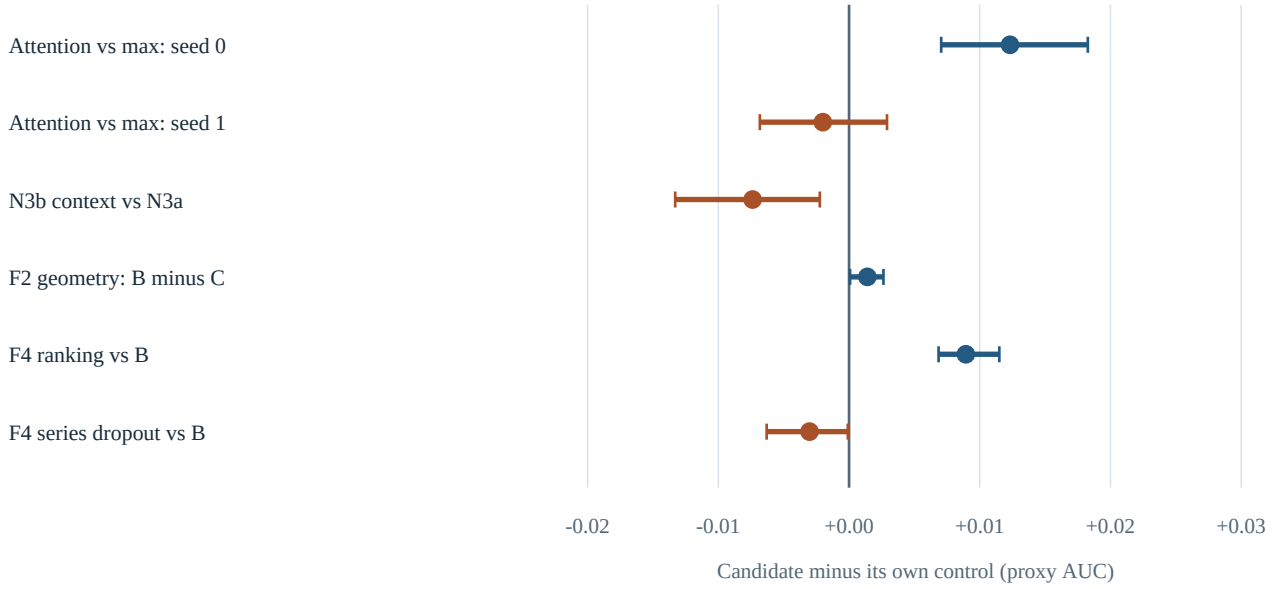


Figure 2. Within-panel paired differences against the stated control. Intervals condition on one fitted run. The context and geometry comparisons have different controls and cannot be interpreted as a common leaderboard.

Comparison	Control AUC	Candidate AUC	Difference [95% group interval]
Attention vs max, seed 0	0.874449	0.886765	+0.012316 [0.007050, 0.018267]
Attention vs max, seed 1	0.884574	0.882570	-0.002004 [-0.006823, 0.002906]
N3b context vs N3a attention	0.888463	0.881085	-0.007377 [-0.013300, -0.002236]

The initial attention gain did not reproduce. Each pair is hardware-matched internally, but the first ran on a laptop and the second on Kaggle; cross-pair variation therefore mixes seed and environment changes. This is evidence against a robust superiority claim from the existing results, not proof that max pooling is universally better. The N3b result concerns a particular context package and training recipe.

5.3 Frozen-feature controls

Head	Proxy macro AUC	Interpretation
A: simple attention	0.780321	Frozen-feature control
B: context, slot, geometry	0.798874	Combined capacity/context/metadata change
C: B capacity, zero geometry	0.797473	Post hoc geometry diagnostic
B plus ranking loss	0.807809	Single-seed exploratory gain
B plus series dropout	0.795851	No gain in this configuration

B minus A was +0.018553 [0.012153, 0.025107]. This does not isolate geometry. B minus C was only +0.001401 [0.000079, 0.002633]. The ranking arm improved B by +0.008935 [0.006851, 0.011496]; dropout reduced it by -0.003022 [-0.006301, -0.000093]. These results reuse the development fold and require paired-seed and independent-reference confirmation.

Feature extraction on a laptop RTX 4050 took 443.25 seconds for 4,290 studies and 166,008 windows. The paired A/B run, including setup/extraction/analysis, took 512.39 seconds. The C diagnostic took 50.78 CPU seconds, and the

concurrent F4 workers took 84.45 wall seconds. Zero Kaggle GPU hours were used for F2/F4; local GPU extraction was used. These measurements are not a matched-quality efficiency comparison with the stronger end-to-end readers.

5.4 Independently trained competition submissions

Entry	Main changes	Public AUC	Private AUC
E01	Six-epoch ResNet18 baseline	0.769	0.735
E1	Cache/normalization/mirroring and longer training	0.807	0.788
E4+E5	Report-v3, soft targets and loss changes	0.841	0.821
Ens3	Add E6 and E2 through rank averaging	0.848	0.827
Ens4	Add ConvNeXt-Tiny E8	0.857	0.837

These are historical recipe packages, not an isolated causal decomposition of preprocessing, supervision and architecture. The final private-minus-public difference for Ens4 is -0.020. Local report-proxy validation for Ens4 was about 0.880; that value has a different reference and cohort from either leaderboard. Not every later development model was submitted standalone.

5.5 Public-weight reproduction and adaptation

F0 reproduces goodpjw2008's stack using nartaa's Raptor384/Raptor224, heliosli R6 and the small reader [4]. K1 fine-tunes were highly similar to the Raptor parent, with reported prediction-rank correlations around 0.986-0.988. K3/K4 tested approximate compartment tiles; T1/S1 explored teacher-derived targets. Several arms suffered fp16 overflow or collapsed after large encoder updates. These failures were not repaired into a fair comparison for every backbone.

Entry	Change	Public AUC	Private AUC
F0	Unchanged upstream stack	0.958	0.942
F1	F0 plus fixed 20% S1-B student rank blend	0.954	0.938
F3	Shrink main family weights toward equal	0.957	0.942
F5	Remove small reader	0.958	0.942
F6	Gaussian-rank main aggregation	0.958	0.942
F7	Squared percentile ranks	0.958	0.942

All leaderboard values use Kaggle's three-decimal display. Equal displayed scores do not establish equal predictions or unrounded AUC. There was no demonstrated private improvement over F0. S1-B's development gates compared it with a parent on a small expert set, not an honest held-out full-F0 ensemble; its deployed blend regressed. This illustrates an inadequate promotion test without proving a single explanation for the regression.

6. Relation to published competition solutions

The first-place write-up currently presents one teammate's component. It uses agent-produced soft labels, anatomical ROI and lesion boxes, a learned detector, ROI-restricted pooling, and lesion/report-embedding auxiliary tasks. These add supervision and spatial evidence we did not build. The author's single-model efficiency entry reports private AUC 0.954; this is not the complete winning team's ensemble score [5].

Third place reports 0.959 private with 36 checkpoints, image-derived soft targets, depth context, backbone/resolution diversity and group-budgeted probability fusion. OAI training is used in 23 members. Non-refit and expert-case refit variants both display 0.959 private, so their table does not establish a refit gain at that precision. Several interventions are bundled [6].

These primary author accounts suggest promising directions but do not prove which intervention would close our gap. Attention MIL [2] and DINO-plus-Transformer volume processing [3] predate this work. Our report's contribution is empirical comparison and transparent failure/provenance accounting rather than a claim to have invented these components.

7. Lessons, limitations and next experiments

First, better agreement with report-derived labels is an incomplete development objective. We did not implement full-dataset image-first supervision, lesion annotations or a landmark-based single-compartment classifier. It is plausible that supervision and spatial evidence constrained us, but our experiments do not identify a hard label-quality ceiling or causally explain the full gap.

Second, a significant-looking interval for one trained run does not establish repeatability. The attention replication is a direct warning against selecting an architecture on one seed. Context/capacity controls are also necessary: the F2 A/B difference was much larger than the isolated geometry diagnostic.

Third, a diverse model is not automatically a useful ensemble member. Prediction correlation is not error correlation. Development gates should measure the paired change of the complete proposed ensemble under an honest reference. Public-parent training on available development studies complicates such a reference; making a new random split does not erase that history.

Fourth, compute and deployment engineering influence which comparisons are actually feasible. Cache parity, native spatial detail, series boundaries, laterality, masks, source snapshots, fail-fast numerical checks and decoding costs are part of a reproducible experimental method. A three-visible-study preview establishes execution, not full hidden-runtime feasibility.

The study is limited by weak targets, adaptive experimentation, limited paired seeds, changing cohorts, unresolved patient/site-independence checks and historical use of the small expert set. F2's large cache-array hashes were retained from recorded provenance rather than recomputed during this audit. E01 saved all six epochs but has a documented late summary-status inconsistency. Numerical failures and incomplete follow-ups are not successful comparisons.

The highest-value confirmation would freeze one question, repeat the cheap geometry/ranking heads across paired seeds, and repeat the attention pair in one environment. Stronger diagnostic claims require an independent reference and a frozen design; the consumed expert set and private leaderboard cannot serve as newly untouched tests. This report launches no confirmation training.

8. Reproducibility and availability

The report is accompanied by machine-readable aggregate results, an evidence manifest, vector figures and a deterministic report builder. The builder reads only those aggregate files and report text. It does not access MRI, labels, per-study predictions, credentials, model checkpoints or network services. The package is therefore reproducible as a report; full training reproduction requires separately obtained data and exact recorded environments/artifacts.

Relevant repository source includes canon.py, m00_model.py, v3_model.py, train_v3.py, f2_data.py and f2_model.py. The report makes no new ownership or license claim over upstream code, weights or competition data. Release/access terms remain those of their respective sources. No patient images, identifiers, report text, label rows or model binaries accompany this report.

9. Conclusion

Our strongest independently trained scored entry achieved 0.837 private AUC, while a reproduced public stack achieved 0.942. Within explicit weak-label protocols, pretraining helped, broader fixed-budget sampling did not, and an initial attention advantage failed to replicate. Frozen-feature experiments identified cheap hypotheses for confirmation but did not establish clinical or hidden-test gains. The reusable outcome is a documented model-selection process with controlled comparisons, negative results and clear provenance.

References

[1] RSNA Knee Abnormality Detection competition. Official task, cohort and evaluation.

[2] Ilse, Tomczak and Welling. Attention-based Deep Multiple Instance Learning. ICML, 2018.

[3] Mueller-Franzes and colleagues. Medical Slice Transformer. Preprint 2024; Scientific Reports publication linked from the paper record.

[4] goodpjw2008. Public CoAtNet/R6/2.5D MIL stack. F0 reproduced version 27, with weights and code attributed to nartaa, heliosli and the upstream reader authors.

[5] Team Kangaroo, teammate's component. First-place solution. Author write-up, accessed 25 October 2026.

[6] DataTech team. Third-place solution. Author write-up, accessed 25 October 2026.

Appendix A. Internal evidence map

Machine-readable source hashes are listed in data/evidence_manifest.json. Paths below are relative to V2_new_research unless otherwise stated.

Report result	Source record
M00/P00 pretraining	research/P00_P01A_FULL_RESULTS_AND_NEXT.md
D01/S01	research/D01_S01_RESULTS_AND_TWO_PLANS.md
Attention seed 0	research/N1_RESULTS_AND_N3A_DECISION.md
Attention seed 1	research/N6_RESULTS.md
N3b context package	outputs/n3b_context/vs_n3a_public.json
F2 controls	research/F2_LOCAL_DINO_RESULTS_2026-10-20.md
F4 ranking/dropout	research/F4_CPU_TRIALS_2026-10-21.md
K/T/S adaptations	research/K1_F0_PUBLIC_WEIGHTS_TRACK.md
Ens4 implementation	Repository docs/OUR_BEST_MODEL.md; original source code
Final scores	Kaggle account history, checked 25 October 2026

Some aggregate sources are intentionally local and are represented by hashes rather than redistributed artifacts. Their absence from the repository is explicit; this report does not claim a fresh clean-environment reproduction.